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| 7.01  | Graphs of equations and functions   |                                                                                                                               |                                                                                                                                                                                                                          |                                                                                               |             |
| 7.01a | x- and y-coordinates                | Work with x- and y-coordinates in all four quadrants.                                                                         |                                                                                                                                                                                                                          |                                                                                               | A8          |
| 7.01b | Graphs of equations and functions   | Use a table of values to plot graphs of linear and quadratic functions.<br>e.g. $y = 2x + 3$<br>$y = 2x^2 + 1$                | Use a table of values to plot other polynomial graphs and reciprocals.<br>e.g. $y = x^3 - 2x$<br>$y = x + \frac{1}{x}$<br>$2x + 3y = 6$                                                                                  | Use a table of values to plot exponential graphs.<br>e.g. $y = 3 \times 1.1^x$                | A9,<br>A14  |
| 7.01c | Polynomial and reciprocal functions | Recognise and sketch the graphs of simple linear and quadratic functions.<br>e.g. $y = 2$<br>$x = 1$<br>$y = 2x$<br>$y = x^2$ | Recognise and sketch graphs of: $y = x^3$ , $y = \frac{1}{x}$ .<br>Identify intercepts and, using symmetry, the turning point of graphs of quadratic functions.<br>Find the roots of a quadratic equation algebraically. | Sketch graphs of quadratic functions, identifying the turning point by completing the square. | A11,<br>A12 |
| 7.01d | Exponential functions               |                                                                                                                               |                                                                                                                                                                                                                          | Recognise and sketch graphs of exponential functions in the form $y = k^x$ for positive $k$ . | A12         |

| GCSE (9–1)<br>content<br>Ref. | Subject content         | Initial learning for this<br>qualification will enable<br>learners to... | Foundation tier learners<br>should also be able to... | Higher tier learners should<br>additionally be able to...                               | DfE<br>Ref. |
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| 7.01e                         | Trigonometric functions |                                                                          |                                                       | Recognise and sketch the<br>graphs of $y = \sin x$ , $y = \cos x$<br>and $y = \tan x$ . | A12         |
| 7.01f                         | Equations of circles    |                                                                          |                                                       | Recognise and use the<br>equation of a circle with centre<br>at the origin.             | A16         |